Autonomous Navigation using Vision-Based Localization and A* Path Planning

Objective: Given a camera that is recording the map, enable a robot to autonomously determine its position, identify goals, and navigate collision-free paths using computer vision and classical path planning algorithms.

Authors: Luigi Tisci, Pietro Bernardo Maria Forni, Matteo Vitali

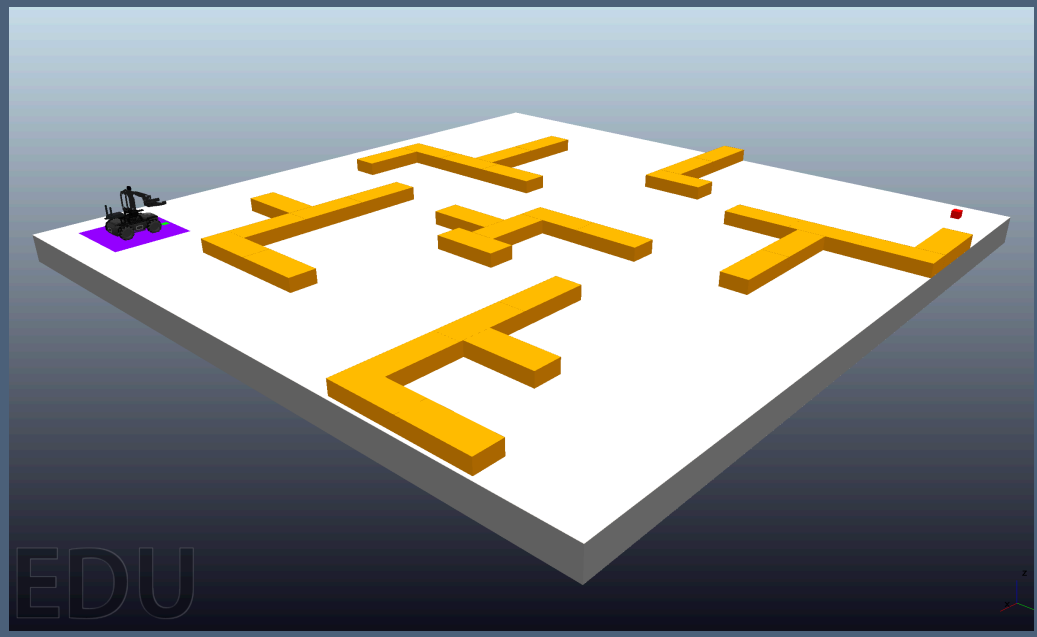
System Architecture & Methodology

Four Main Components:

1. **Camera System**
   - Fixed overhead camera for top-down environment view
2. **Map Processor**
   - Binary occupancy grid generation with safety margins
3. **Path Planner**
   - Optimized A* algorithm
   - Path optimization through line-of-sight simplification
4. **Trajectory Controller**
   - State machine controller with 5 operational phases
   - Conversion from map position to real world position

Baseline Implementation

Setup:

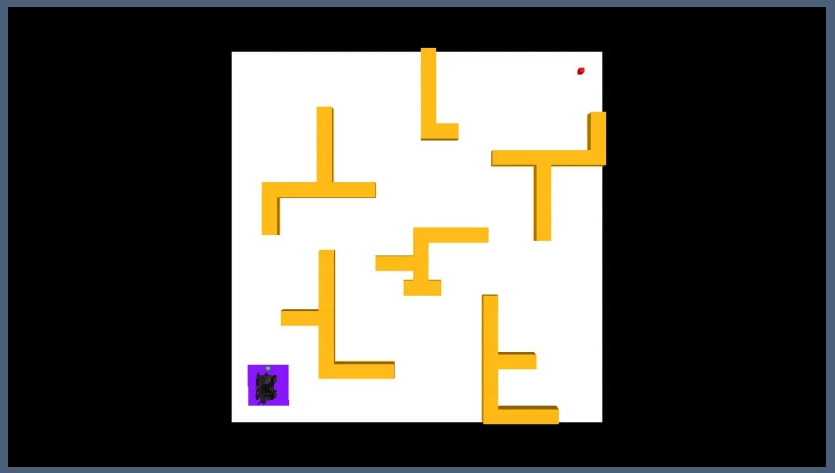


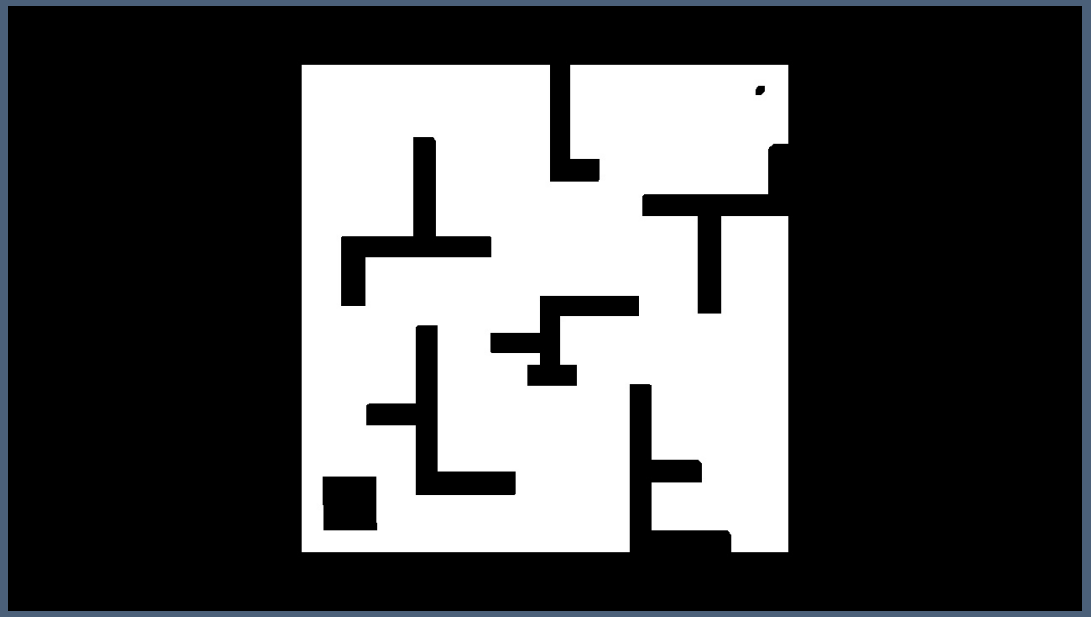
Key Features:

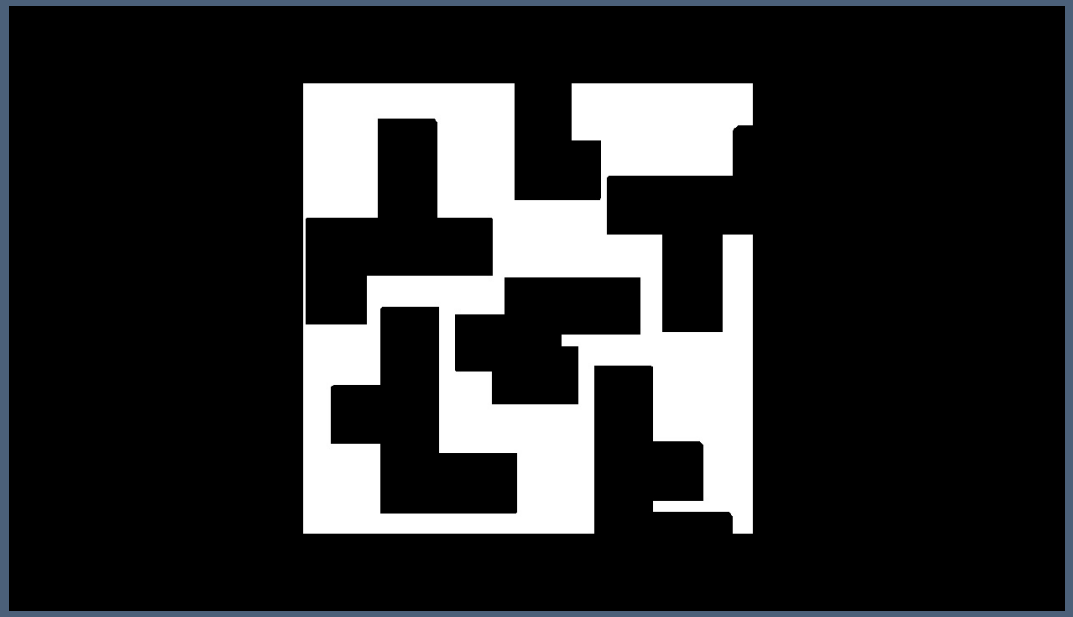
- Single image acquisition at t=0
- Conservative velocities (0.3 m/s linear, 0.1 rad/s angular)
- Time-based rotation control

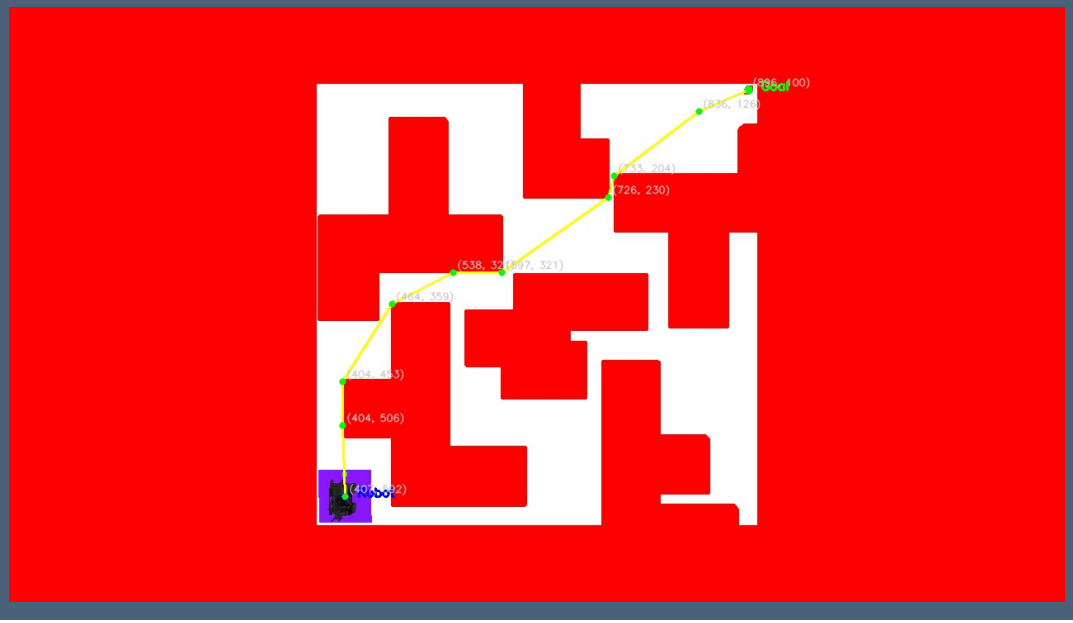
Results:

Reliable performance with conservative parameters, but fails with higher velocities due to accumulated orientation errors.





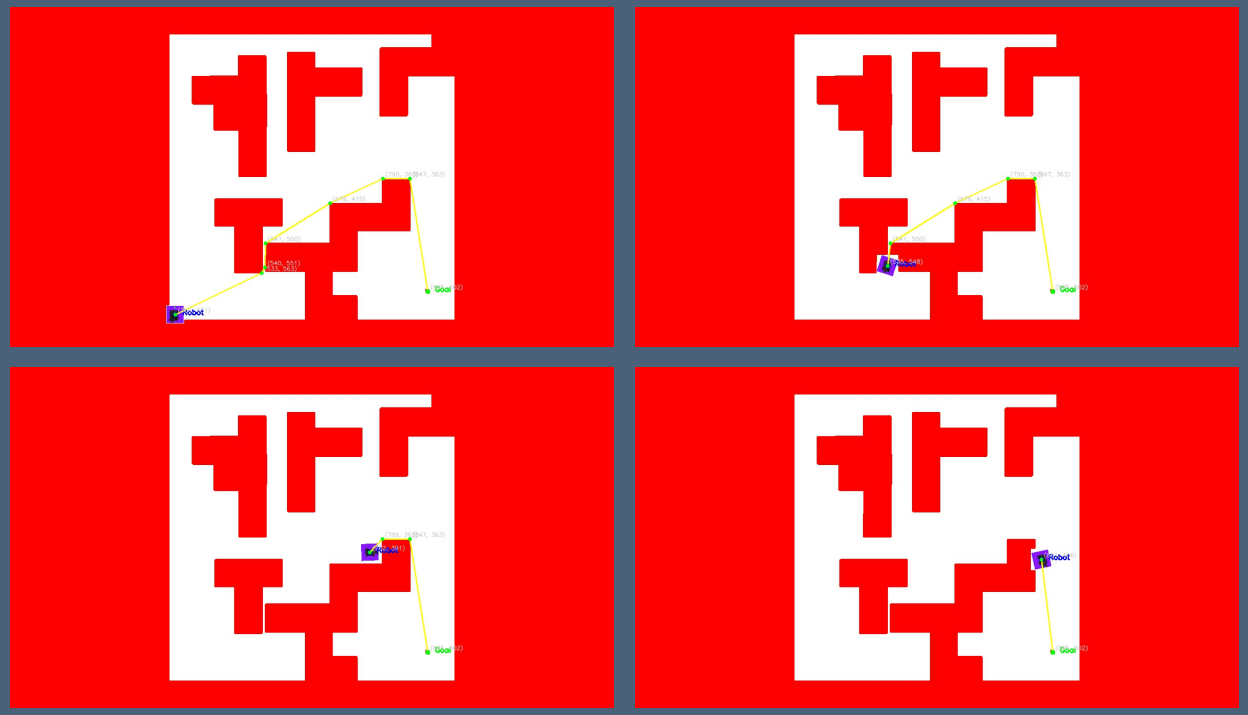



(726, 230)
(404, 506)

Advanced Implementation - Multiple Snapshots

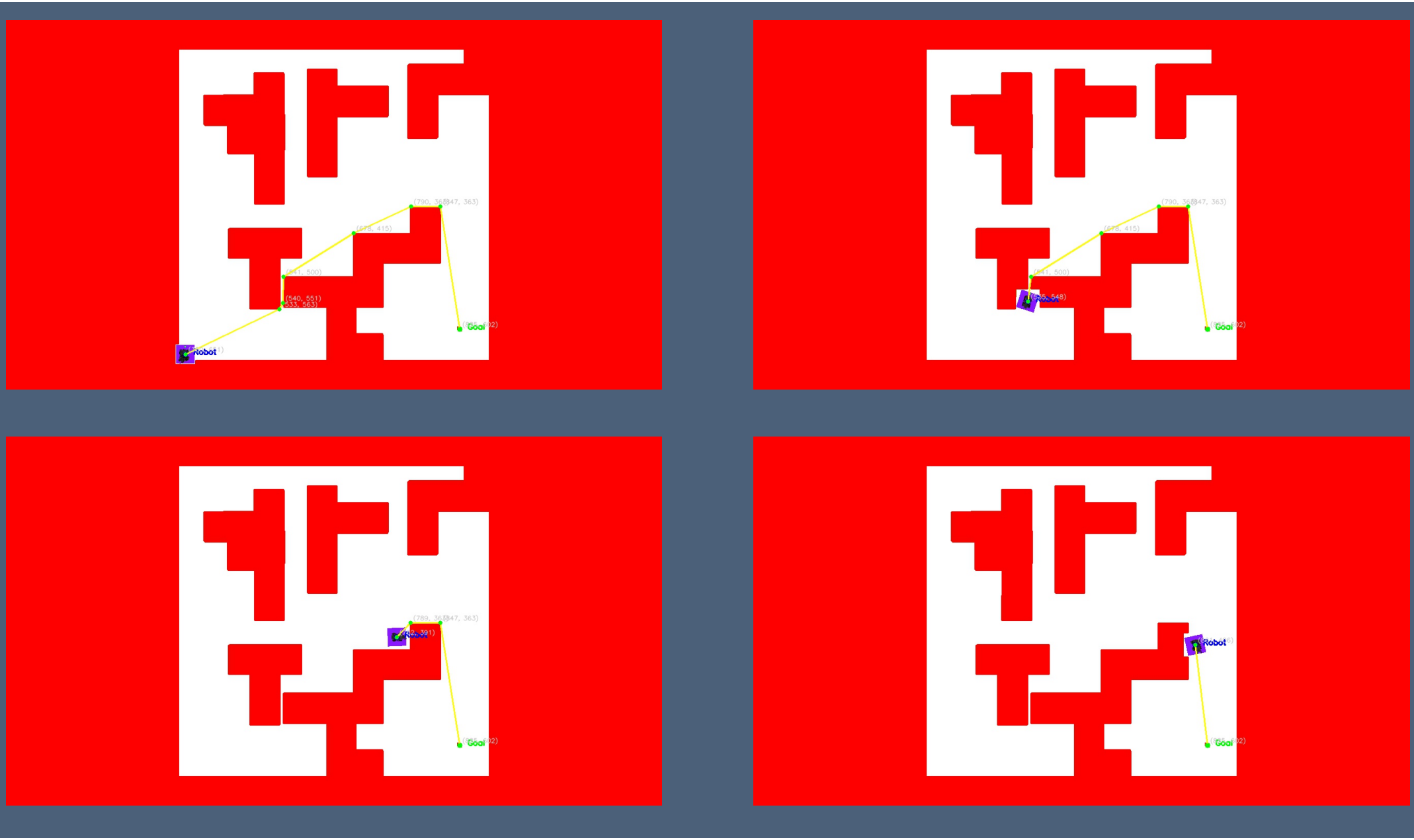
Improvements Made:

1. **Problem Solved**
   - Eliminated accumulated localization errors
   - Reduced trajectory drift at higher speeds
2. **Enhanced System**
   - Periodic map snapshots at fixed intervals
   - Robot stops during image capture



Key Benefit:

- Maintains accuracy throughout the entire navigation sequence by continuously updating robot position and path.


(790, 363)
(678, 415)
(541, 500)
(540, 551)
(533, 563)
Robot
Goal

Dynamic Environment Navigation

Advanced Capability:

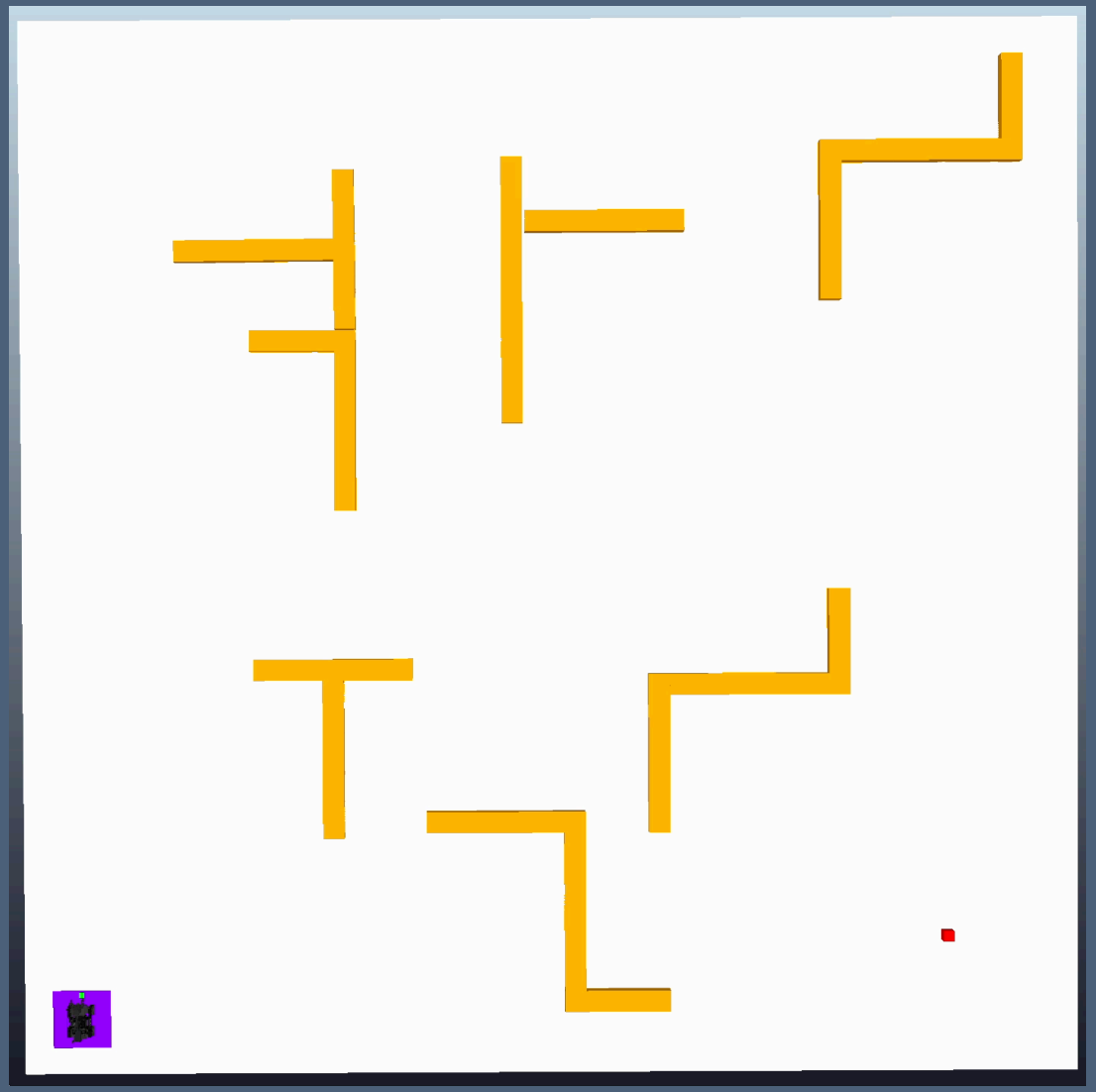
- Successfully demonstrated navigation in changing environments

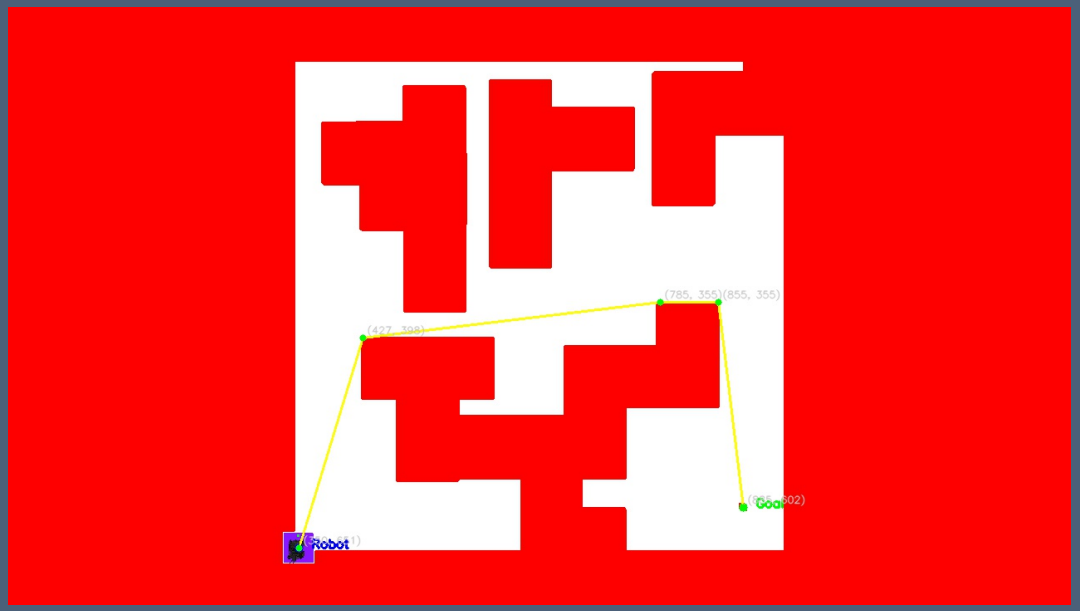
Technical Achievements:

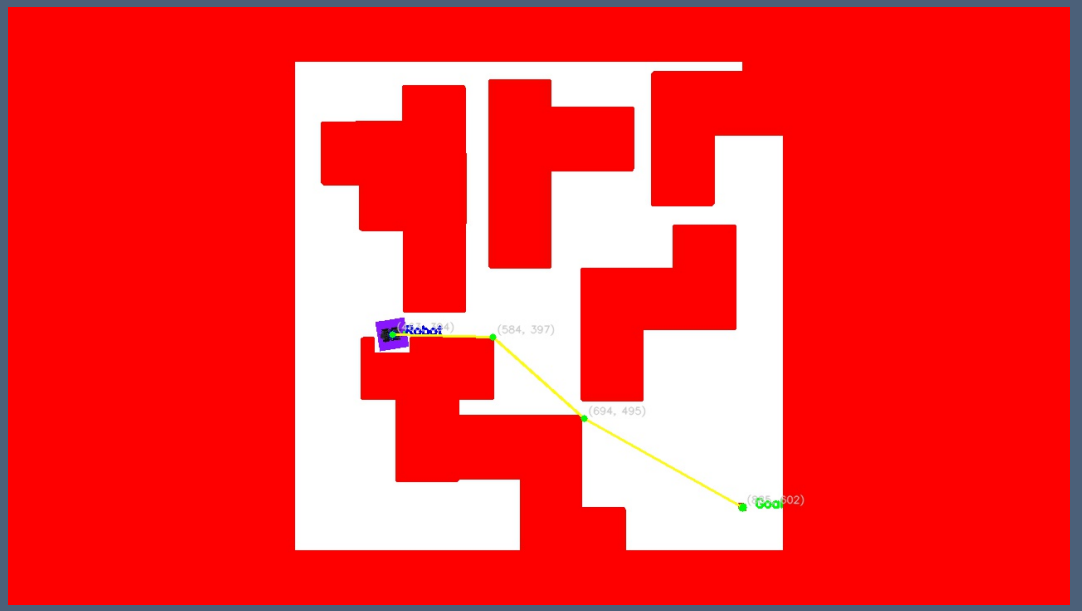
- Constraint: Obstacle movement speed must be manageable relative to update rate

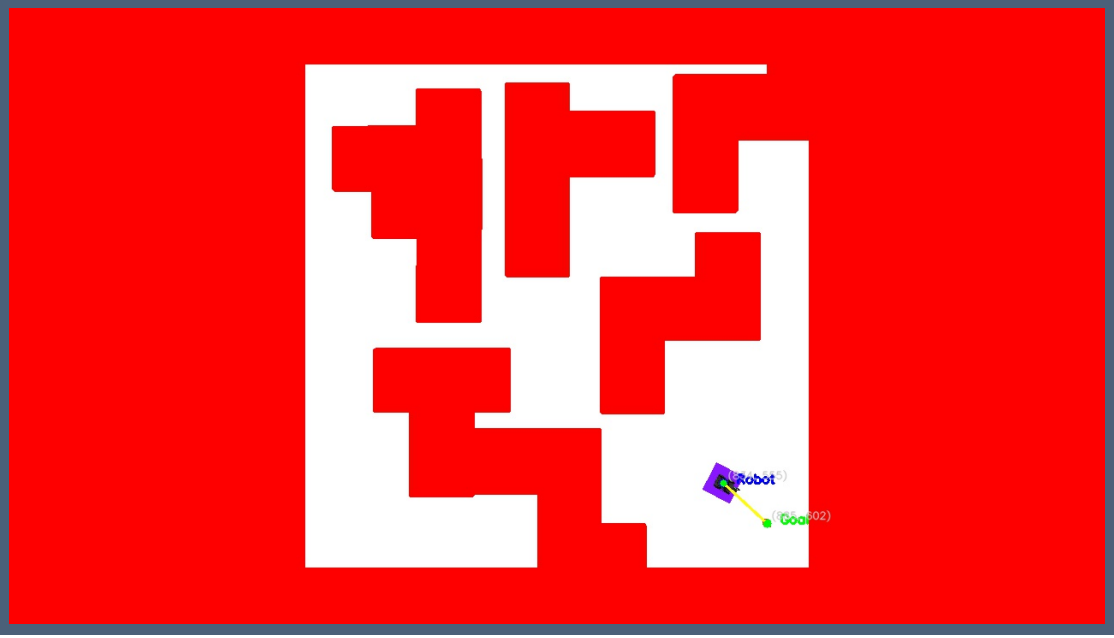
Innovation:

- System adapts to environmental changes by re-processing maps and updating paths in real-time.




(785, 355)(855, 355)
(427, 398)
Robot


(584, 397)
(694, 495)


Robot

Results & Future Works

Technical Limitations:

- The scene parameters (margin, angular and linear velocity) are experimentally tuned.
- It works only in quasi-2D environments. If the walls are high, perspective causes issues.
- With a higher image retake frequency, speed could be improved. This is another parameter that requires manual tuning.
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